The Innovation Strategy and Sustainable Competitive Advantage in Toei Animation's 3DCG Transformation

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Abstract

The traditional hand-drawn studios are subject to technological and organisational pressures due to the rapidly growing demand for animation worldwide, in addition to the increased copyright reuse and the multi-platform distribution. Based on the three theories, namely dynamic capabilities theory, technology – organization complementarity theory and the organizational ambidexterity theory, this is a single case study study on how Toei Animation's use of 3DCG can provide sustainable competitive advantage. The firm has not just replaced 3DCG for hand drawn work. Rather, it has integrated proprietary 3DLO plugins, 3D planning and production tools from Unreal, 3D backgrounds, and hybrid 2D/3D pipelines into its existing planning–production–marketing–licensing framework. Advantage is the combination of hand-drawn art assets, digital assets and in-house tools, organizational absorptive capacity, an extensive IP catalog, and a commercial network all over the world, not just a single software package. SWOT, STEEPLE, and Ansoff analyses map the internal and external circumstances, highlight the constraints in quality control, talent reskilling, tool governance, and coordinating with other countries, and outline practical responses. The paper provides an account of a process-oriented approach and practical proposals for existing animation businesses facing digital transformation.

Key words: Toei Animation, 3DCG Transformation, Innovation Strategy, Dynamic Capabilities, Sustainable Competitive Advantage

1. Introduction

1.1 Research Background

According to data from the Association of Japanese Animations, the overall anime market amounted to 3.8407 trillion yen in 2024. Revenue generated by commercial animation producers accounted for 466.2 billion yen [1] while sales outside of Japan were nearly 56.5 percent of the total. This discrepancy between the spending of the final consumers and the amount of money that reaches the producers is significant. Despite increased spend, many studios still experience limited capacity, inconsistent value and output, and challenge to retain

value. In that context, 3DCG is not so much a tool that can add depth to a scene, but one that can help organize scenes, cameras, assets and layout to be shared and reused. So long as production relies on highly-skilled hand labor, fragmented outsourcing, and project-by-project scheduling, any increase in demand will probably exacerbate quality fluctuations and bottlenecks [2].

Toei Animation has been a company with a combined planning, production, and marketing system and a "one-idea, many-uses" principle since 1956 [3]. If the image of the digital asset remains constant and it can be used in a variety of projects, the cost of adapting it to become a product, game, live event or even an international campaign decreases. The firm has sufficient capital to be able to continue investing. But with the buying of software, there's no automatic guarantee of longevity. The practical significance of studying the concrete steps in which a mature studio incorporates the digital instruments into everyday production thus becomes apparent: when technology stops being a cost factor and becomes part of the organisation's capacities.

Toei built its reputation on hand-drawn 2D imagery, which had been the defining characteristic of Japanese animation for a long time. In recent decades, however, screen technology has progressed tremendously throughout the world. As pipelines have become more stable and digital assets are able to be used over and over, 3DCG has become more appealing. With foreign studios putting pressure on Toei to produce more full-3DCG titles and the expectations of the audience changing, Toei has slowly increased the number of hybrid and even fully 3DCG films. There are several recent theatrical and series projects which already rely on a lot of space that is constructed digitally [4].

It is sometimes suggested that the transition to 3DCG is unavoidable. Some counter-views warn that over-computing can erode the expressiveness of hand-drawn work, which helps keep people engaged for the long term. After some initial experiments in Dragon Ball Z: Resurrection F, Toei has produced films like Dragon Quest: Your Story and One Piece Film: Red with large chunks of 3DCG. The studio's first feature produced entirely in 3DCG, Dragon Quest: Your Story, split opinion — some people thought that the movement of the characters' faces was stiff, and the character lines were not as detailed and sharp as they'd like, while others praised the scenes of large-scale action that would have been extremely costly to produce in pure 2D. The current debates either focus on technological achievements or are an examination of individual works. Few studies connect technological change,

economic consequences and maintenance of artisanal identity within the same framework of an incumbent firm [5].

1.2 Literature Review

Four strands of research are particularly relevant to Toei's context: the evaluation of hybrid production, dynamic capabilities, organizational ambidexterity and digital creative labour.

Sakamoto explored the use of hand drawn animation with 3DCG in Japanese studios. Hybrids were frequently considered by production staff as a valuable alternative for scenes which could not be dealt with efficiently by pure 2D. Mixed style films were often ranked lower, according to audience ratings, than traditional 2D films. The lesson learned here is that efficiency is not all that is required for success but recognizability, production reliability and the ability to work together between the old and the new systems are also important [6].

The dynamic-capabilities approach of Teece, Pisano and Shuen emphasizes the ability to see, do and change. Later studies show positive correlation of digital competencies with those higher order skills, and that the development of these higher order skills leads to a long-term competitive advantage. For Toei, the purchase of 3D software and building 3D libraries are just preliminary steps to resources. For this advantage to endure, a continuous process of recognising changes in audience taste, testing out tools, team reorganisation and control of digital assets needs to be repeated [7]. Capability building is usually interpreted as an ongoing strategic renewal process and not an isolated acquisition [8].

This paper presents results from a study of digital innovation labs across nine organisations, revealing that exploration and exploitation must go hand in hand to create successful digital change. One approach is to move staff members to a lab and then back to the main organisation to ensure that technical results are reincorporated into normal value-creation processes. In an animation company, trials with 3D assets, engine-based layout and AI-assisted tools are considered to be exploration, while schedule discipline, quality inspection and timely delivery are considered to be exploitation. The two activities must continue to be connected [9].

Research on digital creative work offers another take. Tools change interfaces, job boundaries and necessary skills. Automation is often not simply liberation or deskilling, but rather how much autonomy a worker has within the new tool environment. This perspective helps in analysing reskilling and governance within animation production [10].

1.3 Research Gap

The literature tends to focus on the acceptance of a particular technology, the evaluation of a hybrid product, the process of digital transformation, and labour change separately. Usually, efficiency, organizational ability and artistic expression are not studied simultaneously. It is unusual for studies to find a studio's technology, hybrid pipelines, IP system, and world-wide commercial network all within one causal chain that demonstrates how technical advantage can translate into sustainable competitive advantage. In other words, the literature is more apt to answering the question of 3DCG adoption, or whether or not it makes a particular tool more efficient at the local level, than addressing how or whether an incumbent studio can fit 3DCG into its IP operating system and build capabilities that can be reproduced and upgraded. Therefore, it is the aim of the present paper to build an integrated frame and select Toei Animation as the focal case.

1.4 Research Framework

First of all, the actual case of Toei's 3DCG activity is recorded, that is, how Toei's 3DLO plugins, UnrealEngine tools, and 3D backgrounds are adopted in the background modeling, layout, photography, and checking. Practical benefits then come into focus, such as lower costs for spatial reconstruction, more participants can contribute to layout and checking, and more digital assets are linked to the IP value network. Three strategic instruments, SWOT, STEEPLE and the Ansoff Matrix, are used to map out the internal and external contexts, and identify the growth logic of the strategy. Then constraints regarding artistic quality control, talent reskilling, asset governance and cross-border collaboration are identified. The final chapter suggests specific solutions: dual art-technology management, modular trainings and better governance of the global production network. The goal is to get to the heart of the matter and understand under what circumstances sustainable advantage can arise.

The guiding question used is: how does Toei Animation's 3DCG transformation strategy facilitate building sustainable competitive advantage and what most inhibits that facilitation?

2. Case Description

2.1 Toei Animation's 3DCG Evolutionary Trajectory

Not all the titles are in full 3D. Digital solutions have been implemented at specific points in the current pipeline. The move has gone from 3DCG background modeling to internal Unreal Engine instruments to a more general hybrid solution. The key dates are shown in Table 1.

Table 1. The history of Toei Animation's 3DCG Transformation

Period	3DCG Transformation Event	Production and Organizational Impact	Strategic Implication
October 2021	Tropical-Rouge! Pretty Cure the Movie employs 3DLO add-on	No need to redo work later when using 3DCG background models.	First digitisation targeted background, spatial relations and correction stages
April 2023	Run for Money: The Great Mission introduces Unreal Engine internal tools (3DLayout UE)	The participation of external creators or non-CG staff is allowed for model use, layout creation and setting checks.	Technical scope is extended to picture assets, production interfaces, checking processes and on location collaboration.
FY2025	Consolidated net sales of 100.836 billion yen; capital adequacy ratio 80.2 percent; approximately 250 works produced annually	Requires coordination at large volume and various types of titles.	System-level capability is not a function of single-tool procurements, asset reuse and organizational coordination are key factors in enabling system level capability.

The first stages of digitization that the studio used were backgrounds, spatial relations, and correction stages with the 3DLO add-on (as shown in Table 1). Subsequent use of Unreal Engine tools for Run for Money: The Great Mission enabled the use by external creators and non-CG personnel for models, layout construction and setting checks. Time, place and equipment restrictions were thus eased. Initially a tool to store picture assets, this was slowly transformed to a production interface, and a way of collaborating on site [11]. Previous real-time rendering experiences with PreCure series had already shown that the Unreal Engine could achieve a quality comparable to offline rendering with careful adjustments to the cel-look pipelines [12].

2.2 Technical Architecture and Production Integration of Hybrid Pipelines

The working model can be defined as “3D-assisted background—2D character—engine-based layout.” Movable and reusable spatial structures are created by the 3D-assisted background; hand-drawn 2D characters ensure brand recognition and drawing performance; and the IP and license system decides if the assets can migrate between media. 2D/3D integration, fully digital pipelines and global IP management are part of a single business model, in corporate statements anyway. Style recognition and operability of digital-asset are thus considered as complementary instead of competing assets. Real system-level capability hinges on cross-project standards, asset reuse and organizational coordination rather than the purchase of any particular software package, as Toei coordinates many diverse projects annually.

Analysis of Strategic Implications and Constraints: 3.

Toei Animation's 3DCG Transformation – 3.1 Operational and Commercial Value

3.1. 1 Cutting the Costs of Spatial-Reconstruction and Improving Parallel Production Resilience

In film production, the 3DCG background models reduce the amount of subsequent corrections. In production, Unreal Engine tools allow people who don't specialize in CG to manage models, perform checks, and set up layouts. Once there are digital data of scenes, cameras, object position and shot information, adjustments can be made at the parameter level, instead of redrawing frame by frame. If a series uses the same characters and settings again and again, this additional cost of hand drawing and creating the space diminishes.

While the operational advantage is real, it must not be viewed simply as a direct proportional saving in total labour hours. Now with a wider range of skills on the engineering side, there's more of a wider circle of people who can spot problems earlier, meaning that art, direction, photography and production management can intervene earlier. The three-dimensional backgrounds primarily reduce certain aspects of correction, rather than negating the artistic value of hand-drawn performance. Only when low rework rates, reusable assets and earlier collaboration coincide, a durable operational advantage can be achieved. If the naming, the model hierarchy, the version rights and quality standards are not organized, the digitization process can also increase the amount of rework, cause confusion in data and increase maintenance costs [6].

3.1. 2 Technical Capabilities for embedding in the IP Value Network

Toei's combined planning-production-marketing system and one-for-all approach to producing content still hold, and the company continues to focus on earning fees and overseas posts. When characters, world sets, key props, scenes and motion assets are used in the creation of long running IP, reusing them in animation, games, events, merchandise and overseas campaigns can help to minimize the cost of rebuilding. Space and camera control and asset operability via hand-drawn characters and art supervision ensure the visual identity. This allows for this division of labor, which is similar to that of previous results on hybrid products [6].

The benefit isn't as simple as digital imagery being easier to commercialize. But ultimate licensing revenues rely on IP recognition, consumer demand, terms of contracts and local

market conditions, derivative design, and brand protection. What the 3DCG transformation really provides is a technical interface around which various departments can coordinate: production creates controllable assets, rights and marketing departments evaluate the usefulness of the content across platforms and overseas bases organize merchandise, events and communications for regional audiences. The interface should be more competitive than technical, meaning that style governance, asset rights and quality control should be all set-up together, or else it is still a technical interface. Media-mix and platform dynamics add to the returns that can be earned from well-managed digital assets [4].

Strategic Environment Evaluation using Analytical Tools (3.2)

3.2. 1 SWOT Analysis

SWOT breaks down the current 3DCG strategy in four categories: internal strengths, internal weaknesses, external opportunities and external threats.

These strengths include a long-term library of animated IP, proprietary tools like 3DLO and 3DLayout UE, years of in-house CG experience, and the ability to invest in digital spending over a multi-year period with their capital position [3]. The high cost of full 3DCG pre-production projects, the short-term profit pressure in the initial stages of recovery and resistance from experienced hand-drawn artists to new workflows are the weaknesses. High quality animation content that combines 2D and 3D style is a demand of streaming platforms, as well as the growing markets of overseas licensing opportunities [1]. Other studios making investments in CG pipelines is a threat, as well as the lack of younger talent in the traditional hand-drawn space.

One of the drawbacks of SWOT is that it is qualitative. The tool is not able to put a specific numerical value on the financial return of any of the 3DCG projects and it cannot measure cost changes precisely.

3.2. 2 STEEPLE Analysis

STEEPLE is a survey of seven external macro factors that impact the decision to increase 3DCG production.

Demographically, there are fewer young hand-drawn artists to choose from, Japanese animation professionals are getting older and the world is hungry for more variety in anime. In terms of technology, motion capture, AI-assisted rendering, real-time CG compositing are well developed enough to produce 3D images that still look like 2D cel animation; company

press releases talk of a world-class 2D–3D fused visual system [12]. Economically, pure hand-drawing becomes increasingly labor intensive over the years, while the digital approach is more labor saving in the long term, but requires higher initial investment and training costs, which increase fixed costs in the short-term. Environmentally, digital pipelines reduce the use of art materials. Politically, the government programs promote the reform of working practices and reduce overtime in creative industries. In the legal realm, Japanese cultural policy still shows a preference for digital-content innovation. From that perspective copyright control and the prevention of unauthorized use are still regulatory issues of prime importance.

What STEEPLE also makes clear is that it only considers the outside picture. It does not encompass internal opposition from top artists or wrangling over budgets between traditional and digital units.

3.2. 3 Ansoff Matrix Analysis

Ansoff Matrix is a classification based on market and product dimensions. Toei's 3DCG activity basically follows a product-development line: the company stays within the target market of anime fans while developing new technical features, such as hybrid 2D/3D pipelines and engine-based layout to provide differentiated visual products. The strategy is not primarily focused on finding new customer groups. Instead, it's designed to shore up the current fan base and market to viewers who are more interested in the stronger spatial and camera work—while also helping to bolster revenue goals over the long haul.

A drawback of the Matrix is that it focuses on market–product logic and provides little information on the financial risks on project level. Its range of explanation does not include high initial investment and multi-year recovery cycles.

3.3 Listed constrains in Hybrid production and Global Operations

3.3. In this section, the risks affecting the artistic quality are identified and listed.In this section, the risks that affect the artistic quality are identified and listed.

Public materials emphasize 3DLO tools, Unreal Engine instruments, 2D/3D integration, and fully digital pipelines, but reveal virtually no project-level metrics like tool coverage rates, shot rework rates, asset-reuse ratios, or quality scores. Hybrid production is therefore a part of the stated strategy, and stable artistic quality cannot be assumed. Visual continuity and language between materials, line quality, color, lighting, camera moves, animation supervision and digital techs, become very important in maintaining visual continuity when characters are in 2D hand-drawn and backgrounds and photography are in 3D. Tools that

increase coordination costs but do not decrease them will include standardized shot templates, style reference libraries, asset-review gates and problem-traceability routines.

3.3. Two Talent Reskilling Challenges Based on the Changing Job Capabilities

Clearly the tools in Unreal Engine make models easier for non-CG staff to work with and check layouts. The above does not mean that creative people are already three-dimensional, fluent in the engine, skilled in data-management, or accustomed to real-time collaboration. Inside the animation production process, digital tools change the boundaries and skill requirements of occupations. There are two dangers that occur at once when training focuses purely on the use of the software: senior artists lose creative control because they cannot control the software, and junior technicians lose the ability to support the existing aesthetic because they do not have a proper understanding of any hand-drawn principles [10].

3.3. There are three governance issues to consider for cross-border collaboration and digital asset management.

The corporate growth pillars comprise the studio evolution, strengthening of IP, regional expansion and increased contact with customers, along with a medium-term goal of increasing the ratio of overseas sales. Cross-border activity means time-zone differences, language problems, software-license questions, data-security concerns, artistic judgment problems, contracts, and copyright governance. The firm may create more communication nodes without scale if the global production network is used only to outsource low-value stages and if there are no clear guidelines about interfaces, quality, and decision-making between the headquarters and regional teams or with partners. Dynamic-capability research highlights that sensing, seizing and reconfiguring need to be concurrent processes if the investment in the digital is to create a sustainable competitive edge [8].

4. Managerial Recommendations for Hybrid Digital Transformation

4.1 Dual Artistic–Technical Leadership and Reusable Style-Asset Libraries for Hybrid Quality Assurance

For quality-control risks, every major project should have an artistic director and a technical director. The artistic director sets the brand and the boundaries of art, while the technical director is responsible for the arrangement of the assets, their performance, workflow and data management. In the beginning of the project, a “digitizable-style checklist” should be established that defines what is to be digitized, what should stay hand-drawn, and what type of lighting and camera conventions comply with the IP. Art, production, technology and

quality assurance reviews should be incorporated as stage-gate processes. Shot templates, material parameters, naming rules and version records then end up in a controlled asset library, which is reviewed after the project is done. The aim is to transform experimental outcomes into auditable, reusable knowledge in the organization [9].

Modular Capability Training and Cross-Functional Personnel Rotation for Composite Talent Development

Reskilling can be done through capability units that are evaluable and in the form of 2D drawing, art basics, 3D spatial and modeling literacy, real-time engine basics, shot and photography principles, asset and version management, cross-departmental communication, and AI-tool governance. Training is most of the time best, when it is based on real production footage and it is a process of learning, testing, application of the project, review and certification of the process. Senior artists need interfaces to collaborate without losing creative control; junior technicians need training by senior artists; producers need training on asset planning, data permissions and multi-site coordination. The movement of staff within the digital centre of excellence, project teams and overseas collaboration units enables the flow of knowledge across the organization [10].

Capability-Linkage Governance of Global Production Networks and Cross-Border Asset Management

Cross-border cooperation should not be defined as outsourcing, but as capability linkage. Clear interface definition and decision rights are needed with tools, quality standards and partners at the regional level and with external partners. Local teams have the freedom to develop expression for local audiences, but core style and assets are governed by a unified set of rules. Models, materials, shots and rights status should be traceable through a cross regional asset catalog, version-control protocols, data-security and authorization-approval mechanisms. Regional expansion can only result in network-level synergy when there is a common language and common standards which reduce repeated communications [7]. Technical differentiation is fundamental to the success of the business, but if it is not renewed continuously, it can be imitated in a short period of time.

5. Conclusion

5.1 Key Findings

The case of Toei Animation's 3DCG is a clear example of how an incumbent animation

company is using digital technology to restructure hybrid production. Sustainable competitive advantage is not simply the result of 3DCG or AI tools being used alone. This comes from hand-drawn artistic assets plus brand recognition; digital assets plus proprietary tools; production processes plus organizational absorptive capacity; a long-standing IP catalog and licensing experience; and global bases plus capital strength. As per SWOT, STEEPLE and Ansoff analyses, the strategy is based on real strengths and opportunities, but also on some cost pressures and talent resistance, as well as on the threat of competitive imitation. In 3DCG, 3D Construction of Geographical Knowledge, 3D spatial-reconstruction costs are reduced, the possibilities of asset reuse are increased, and the technical interface for cross-border cooperation is enhanced, which supports the operation of IPs. However, commercialization is limited by artistic quality control, reskilling for talent, data governance and rights management. It cannot, therefore, be viewed as just a technical upgrade, but as an integrated art–technology–commerce strategy.

5.2 Research Significance

The study is academically useful because of its linkage of technological investment, organizational governance and IP management. The above described dual-leadership mechanism, style-asset libraries, modular capability training and governance of the global network can reduce the coordination costs that often arise from technical expansion, and can provide a form of institutional support for value extraction from IP assets in the long run. Overall, the analysis highlights the importance of artistic skill in the animation sector, which technology cannot replace. Processes need to be reconfigured and the organization needs to absorb the digital tools before they can generate sustainable competitive advantage.

5.3 Limitations and future study

The study is largely based on available public company documents, statistics, and other literature. Labor-hour savings, cost reductions or financial returns cannot be precisely claimed on the basis of interviews, questionnaires or internal performance data. These (SWOT, STEEPLE, and Ansoff itself) are still qualitative tools, which should not be used in place of the project-level financial modeling. If corporate access can be secured, future research could utilize surveys, interviews, project archives, or comparative cases to provide measurements on net effects of 3DCG, including rework of shots, reuse of assets, project production cycle time, staff skills, project quality, and licensing revenue.

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